Digital Image Processing

Lecture # 13

Texture Analysis & Local Binary Pattern

Texture Based Descriptors

Texture

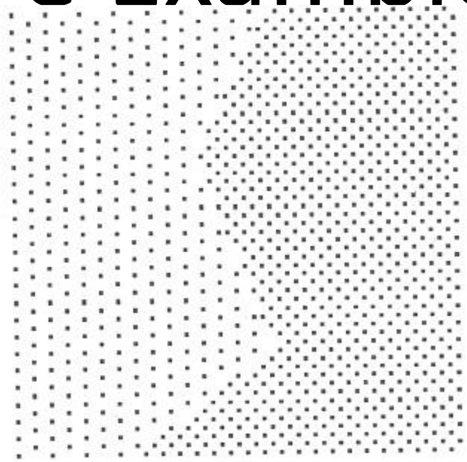
- Organized patterns of quite regular subelements called textons.
- Texture is a property of sufficiently large regions

Applications:

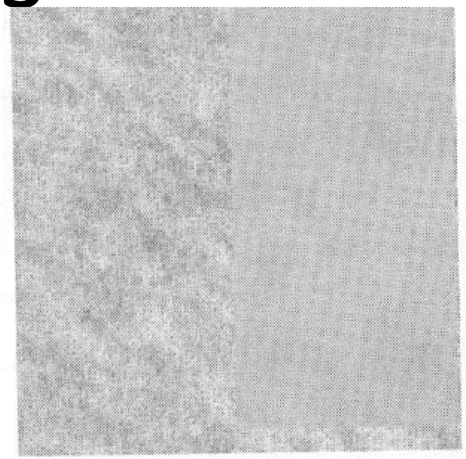
- Texture based segmentation
- Texture synthesis
- Texture analysis and texture based matching
- Shape (surface orientation) from texture

Texture Examples

Test image T1
(a)



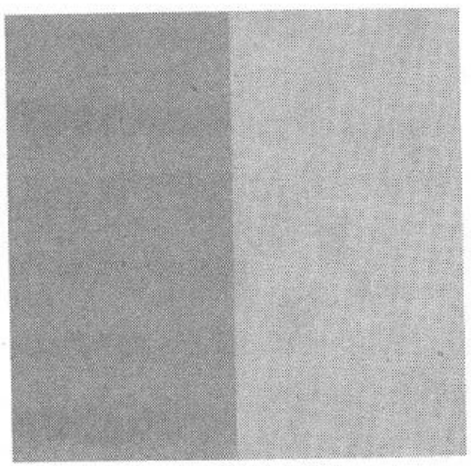
Test image T2
(b)



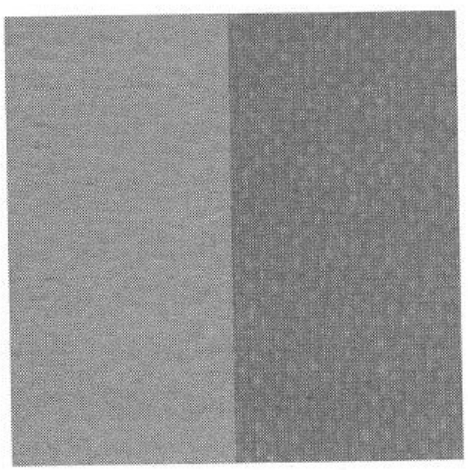
Test image T3
(c)

(a,b): Artificial textures

(c,d,e): Naturally occurring textures



Test image T4
(d)



Test image T5
(e)

Representing textures

Statistical

yields characterization of textures as smooth, coarse grainy, etc.

Spectral

are based on Fourier spectrum and are primarily used to detect the global periodicity in an image by identifying high energy narrow peaks in the spectrum.

Statistical approaches

- Based on the histogram measures of image
- Based on the Grey Level Co-occurrence Matrix (GLCM) and related measurement

Histogram based texture description

Using statistical moments of grey level histogram of the image or region

Let $p(z_i)$ is the histogram of the grey levels z_i of an image

The nth moment about the mean is given by:

$$\mu_n(z) = \sum_{i=0}^{L-1} (z_i - m)^n p(z_i)$$

Where mean is $$m = \sum_{i=0}^{L-1} z_i p(z_i)$$

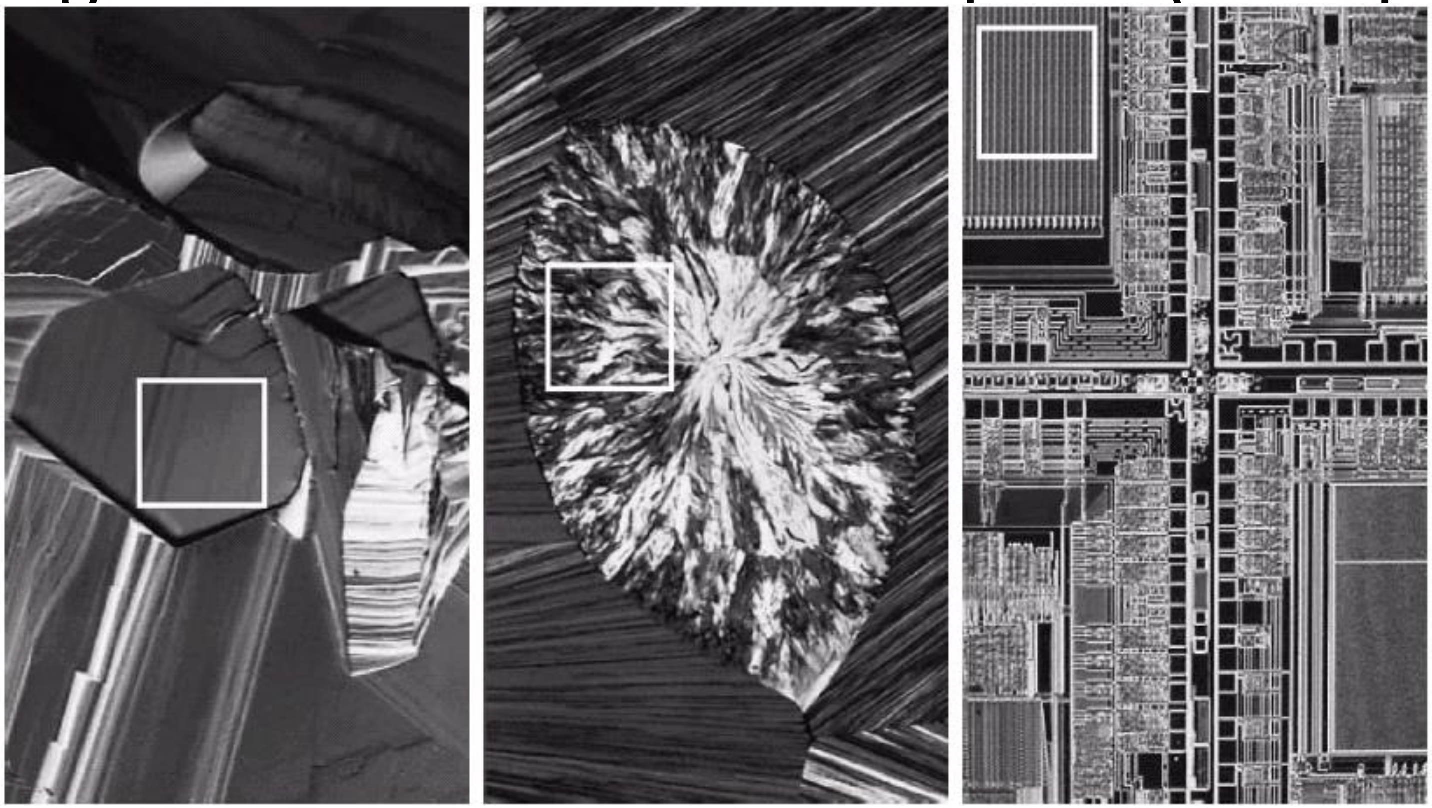
The variance is the second moment and is given by

$$\sigma^2(z) = \mu_2(z) = \sum_{i=0}^{L-1} (z_i - m)^2 p(z_i)$$

Histogram based texture description

- For texture description the following parameters are useful
  - Variance and related measures: descriptor of relative smoothness, use normalized variance $R = 1 - \frac{1}{1+\sigma^2(z)}$
  - Skewness of histogram $\mu_3(z) = \sum_{i=0}^{L-1} (z_i - m)^3 p(z_i)$
  - Relative flatness of histogram $\mu_4(z) = \sum_{i=0}^{L-1} (z_i - m)^4 p(z_i)$
  - Uniformity $U = \sum_{i=0}^{L-1} p^2(z_i)$
  - Average Entropy $e = -\sum_{i=0}^{L-1} p(z_i) \log_2 p(z_i)$

Histogram based texture description (example)



Texture	Mean	Standard deviation	R (normalized)	Third moment	Uniformity	Entropy
Smooth	82.64	11.79	0.002	−0.105	0.026	5.434
Coarse	143.56	74.63	0.079	−0.151	0.005	7.783
Regular	99.72	33.73	0.017	0.750	0.013	6.674

GLCMs

- For texture description the following parameters of GLCM are measured and analyzed

  Maximum probability

$$\max_{i,j}(c_{ij})$$

  Contrast

$$\sum_i \sum_j (i-j)^2 c_{ij}$$

  Uniformity

$$\sum_i \sum_j c_{ij}^2$$

  Entropy

$$-\sum_i \sum_j c_{ij} \log_2 c_{ij}$$

FIGURE 11.29
How to generate a co-occurrence matrix.

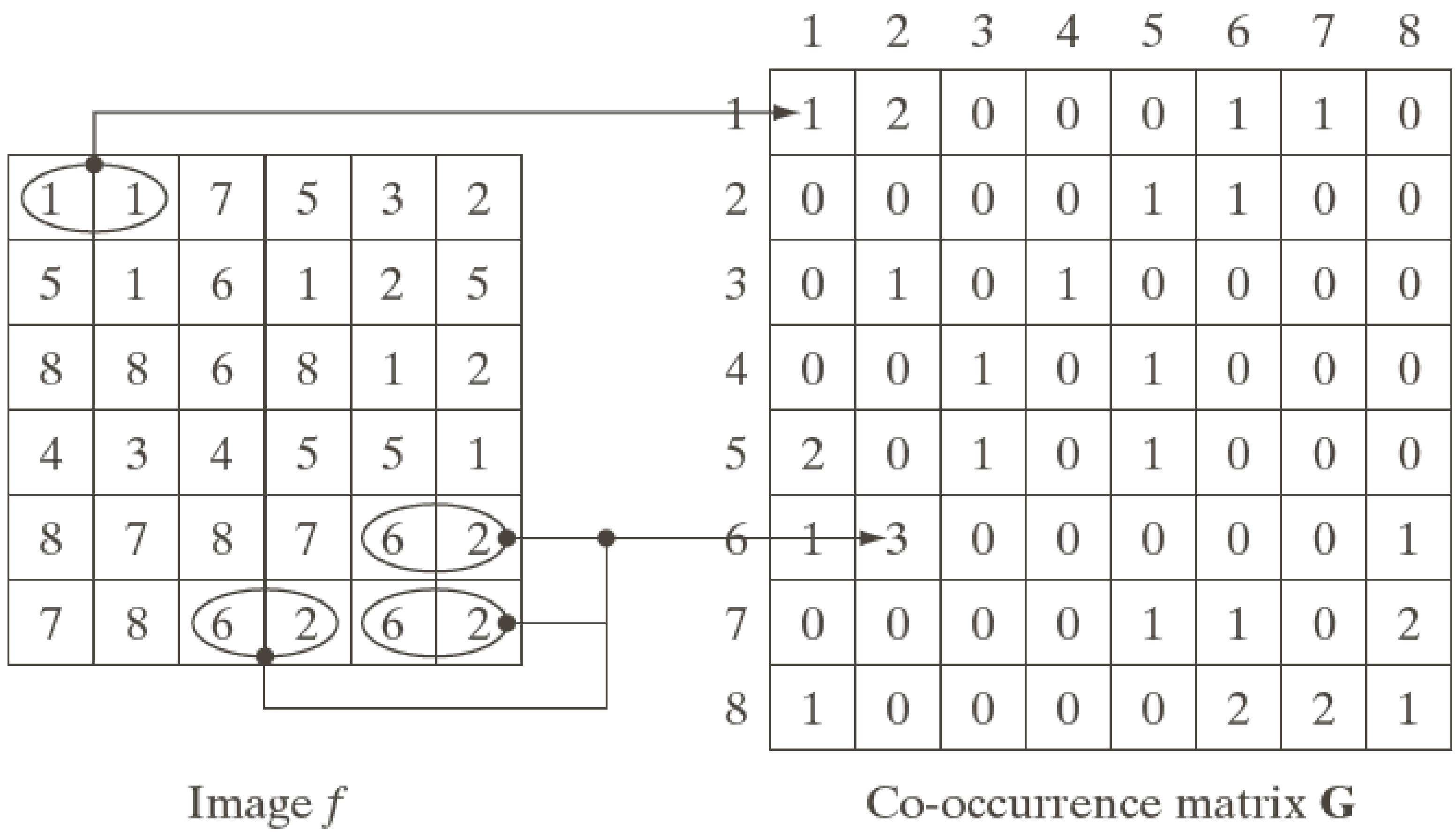
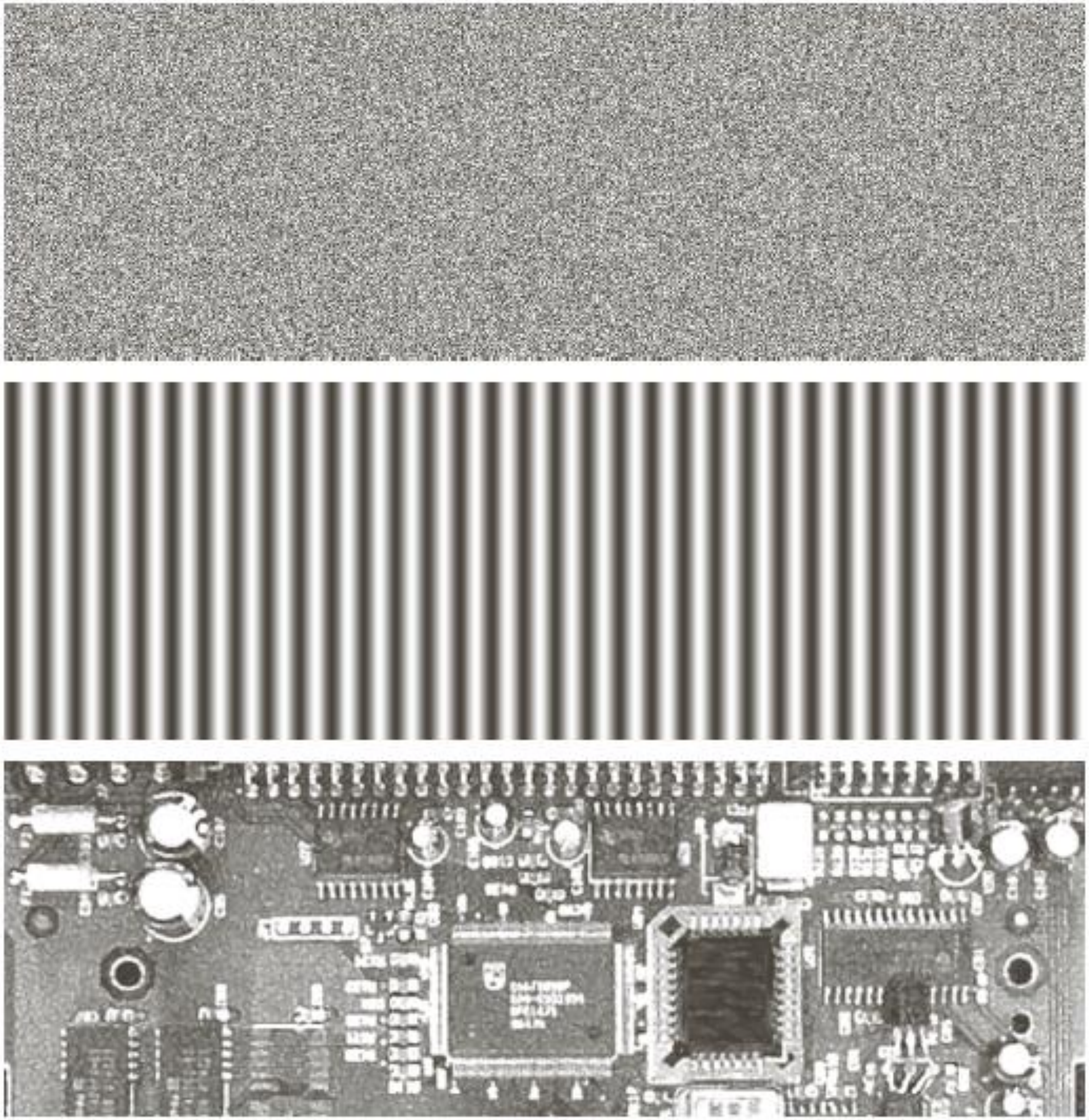


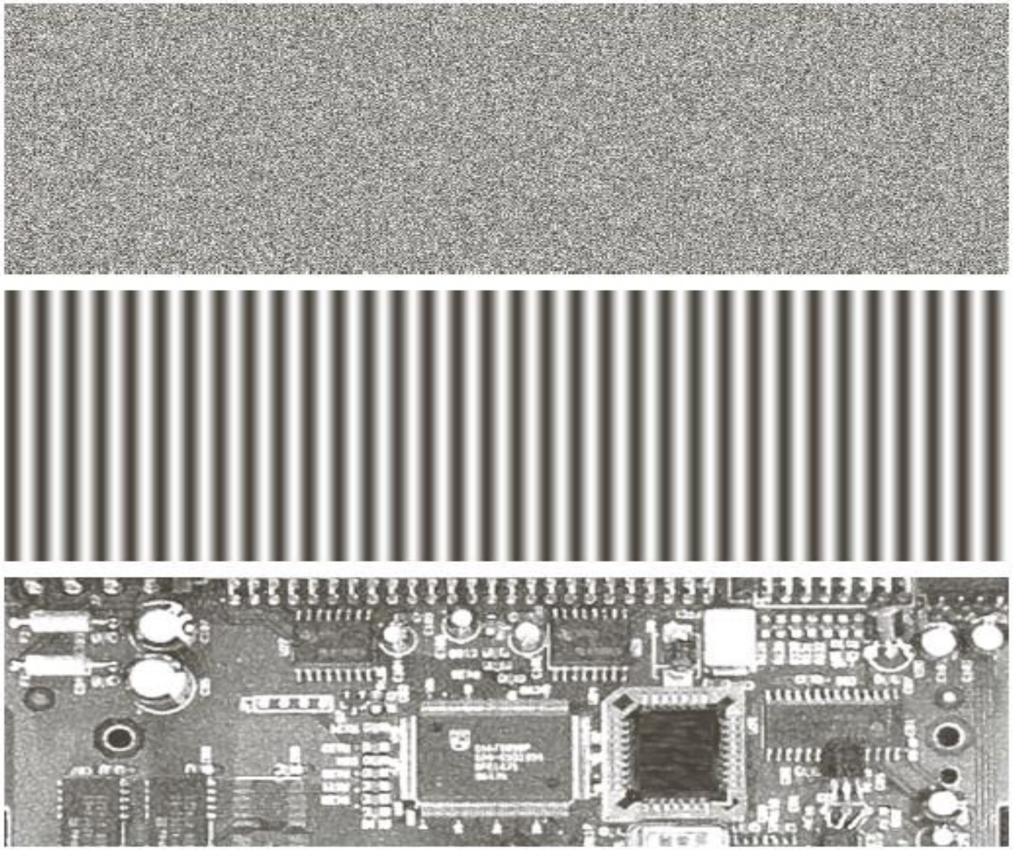
TABLE 11.3 Descriptors used for characterizing co-occurrence matrices of size $K \times K$. The term p_{ij} is the ijth term of **G** divided by the sum of the elements of **G**.

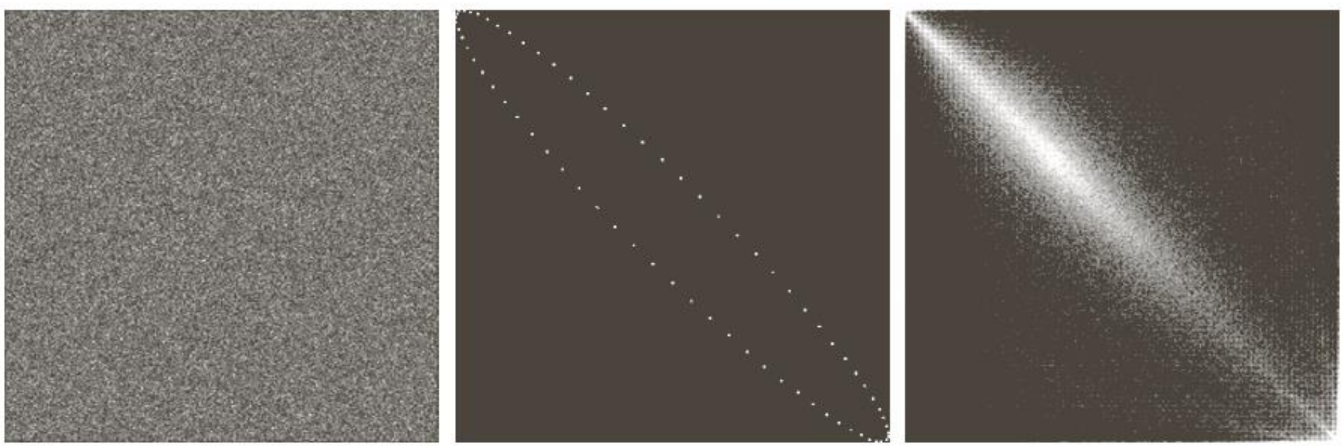
Descriptor	Explanation	Formula
Maximum probability	Measures the strongest response of **G**. The range of values is [0, 1].	$\max_{i,j}(p_{ij})$
Correlation	A measure of how correlated a pixel is to its neighbor over the entire image. Range of values is 1 to −1, corresponding to perfect positive and perfect negative correlations. This measure is not defined if either standard deviation is zero.	$\sum_{i=1}^{K}\sum_{j=1}^{K}\frac{(i - m_r)(j - m_c)p_{ij}}{\sigma_r\sigma_c}$ $\sigma_r \neq 0; \sigma_c \neq 0$
Contrast	A measure of intensity contrast between a pixel and its neighbor over the entire image. The range of values is 0 (when **G** is constant) to $(K - 1)^2$.	$\sum_{i=1}^{K}\sum_{j=1}^{K}(i - j)^2 p_{ij}$
Uniformity (also called Energy)	A measure of uniformity in the range [0, 1]. Uniformity is 1 for a constant image.	$\sum_{i=1}^{K}\sum_{j=1}^{K}p_{ij}^2$
Homogeneity	Measures the spatial closeness of the distribution of elements in **G** to the diagonal. The range of values is [0, 1], with the maximum being achieved when **G** is a diagonal matrix.	$\sum_{i=1}^{K}\sum_{i=1}^{K}\frac{p_{ij}}{1 + \lvert i - j\rvert}$
Entropy	Measures the randomness of the elements of **G**. The entropy is 0 when all p_{ij}'s are 0 and is maximum when all p_{ij}'s are equal. The maximum value is $2 \log_2 K$. (See Eq. (11.3-9) regarding entropy).	$-\sum_{i=1}^{K}\sum_{i=1}^{K}p_{ij}\log_2 p_{ij}$



a
b
c

FIGURE 11.30 Images whose pixels have (a) random, (b) periodic, and (c) mixed texture patterns. Each image is of size 263×800 pixels.





a b c

FIGURE 11.31 256×256 co-occurrence matrices, $\mathbf{G}_1$, $\mathbf{G}_2$, and $\mathbf{G}_3$, corresponding from left to right to the images in Fig. 11.30.

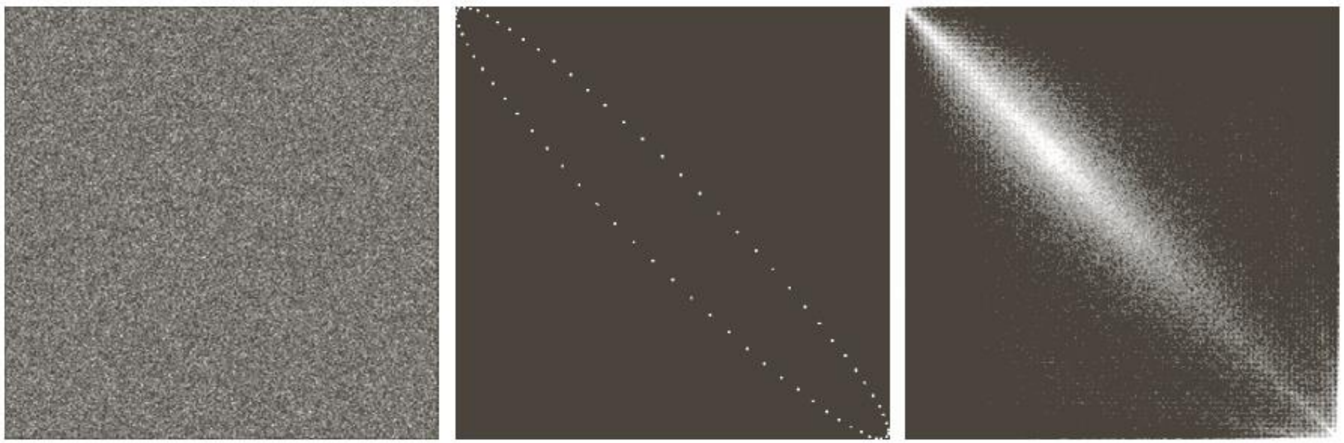


TABLE 11.4 Descriptors evaluated using the co-occurrence matrices displayed in Fig. 11.31.

Normalized Co-occurrence Matrix	Descriptor					
	Max Probability	Correlation	Contrast	Uniformity	Homogeneity	Entropy
$\mathbf{G}_1/n_1$	0.00006	−0.0005	10838	0.00002	0.0366	15.75
$\mathbf{G}_2/n_2$	0.01500	0.9650	570	0.01230	0.0824	6.43
$\mathbf{G}_3/n_3$	0.06860	0.8798	1356	0.00480	0.2048	13.58

Spectral Texture Analysis

Spectral techniques: Fourier transform

- Suitable to detect directionality of periodic and almost periodic 2-D patterns in an image
- Periodic texture patterns are easily detectable by concentration of high energy burst in the spectrum
- Features of Fourier spectrum for texture representation are:
  - Prominent peaks in the spectrum give the principal direction of texture patterns
  - The location of peaks give the frequency and thus the scale of repetition of a pattern
- Eliminating any periodic components via filtering leaves non-periodic image elements which can be described by statistical techniques

Spectral techniques: Fourier transform

- Simplified by expressing the spectrum in polar coordinates to yield a function $S(r, \theta)$ where S is the spectrum function and r and θ are the polar coordinates.

  For each direction θ, $S(r, \theta)$ = a 1-D function $S_\theta(r)$

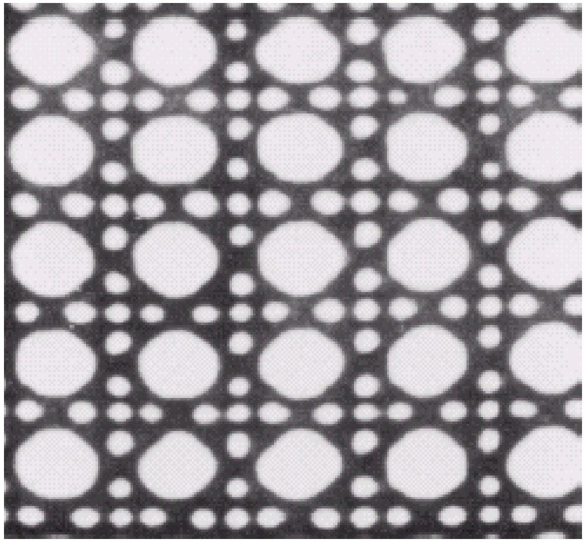
  For each frequency r, $S(r, \theta)$ = a 1-D function $S_r(\theta)$
- Analyzing $S_\theta(r)$ for a fixed θ, gives the distance from the origin and thus the scale of repetition of a texture pattern.
- Analyzing $S_r(\theta)$ for a fixed r, gives the direction and thus the orientation of the periodic texture pattern.
- To measure this analysis, we define two quantities

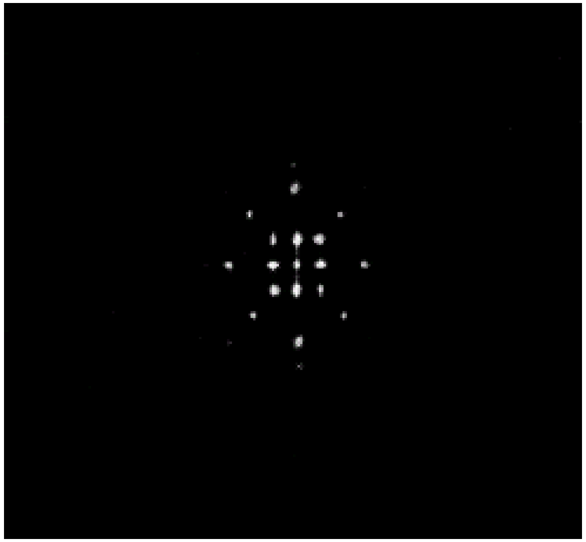
$$S(r) = \sum_{\theta=0}^{\pi} S_\theta(r), \qquad S(\theta) = \sum_{r=1}^{R_0} S_r(\theta).$$

These quantities measure the spectral response and give the dominant directions and scales of periodic texture patterns.

Spectral techniques: Fourier transform

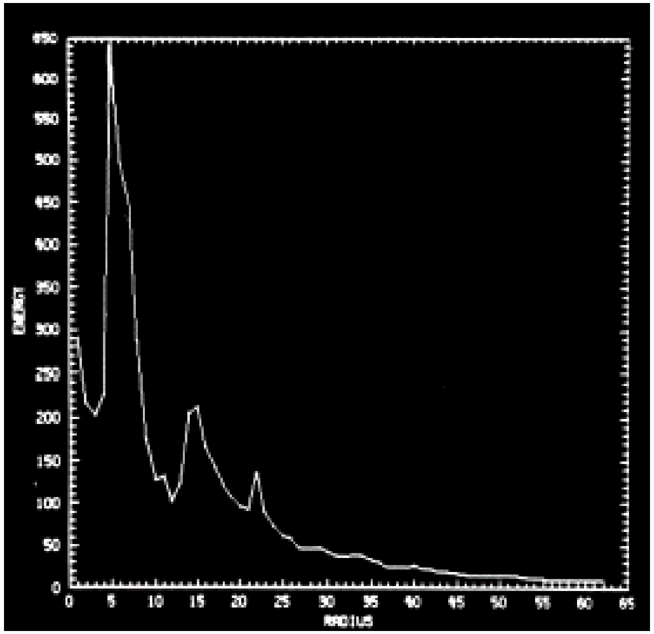
Image showing periodic texture

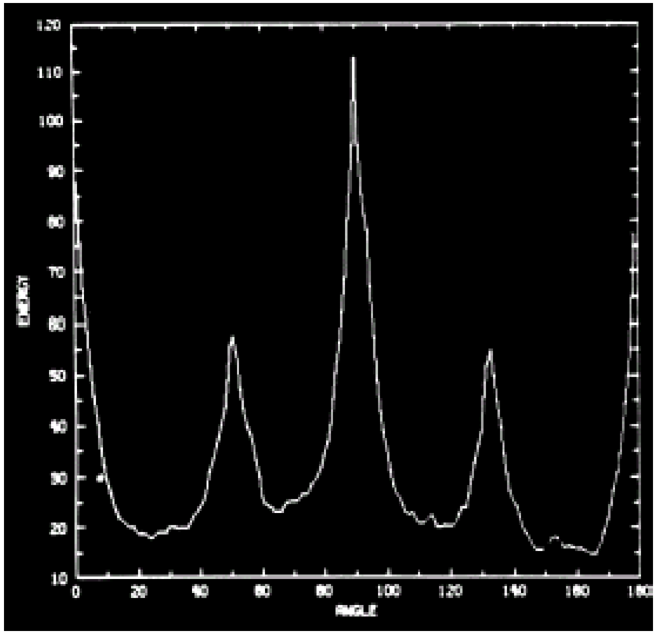




Spectrum

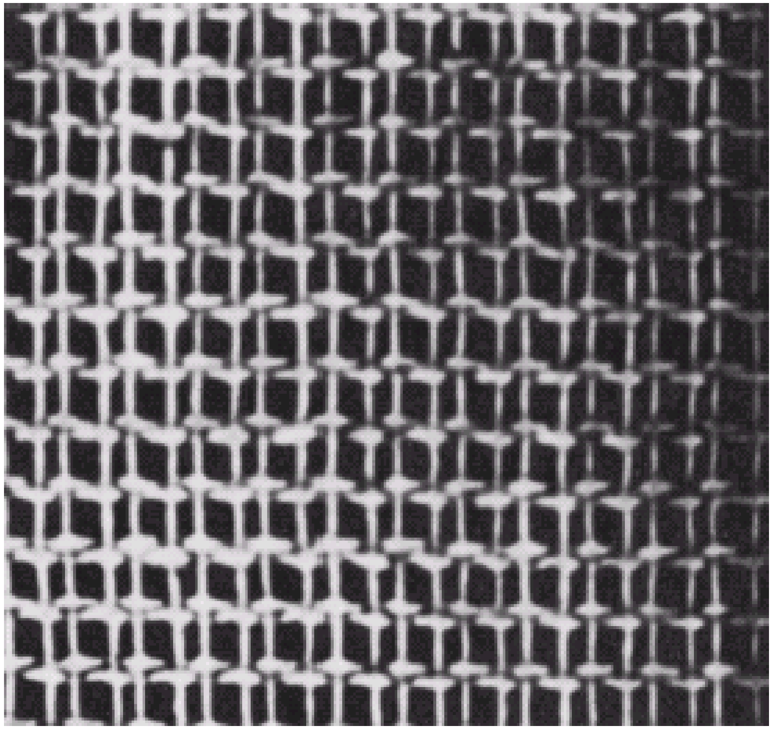
Plot of $S(r)$



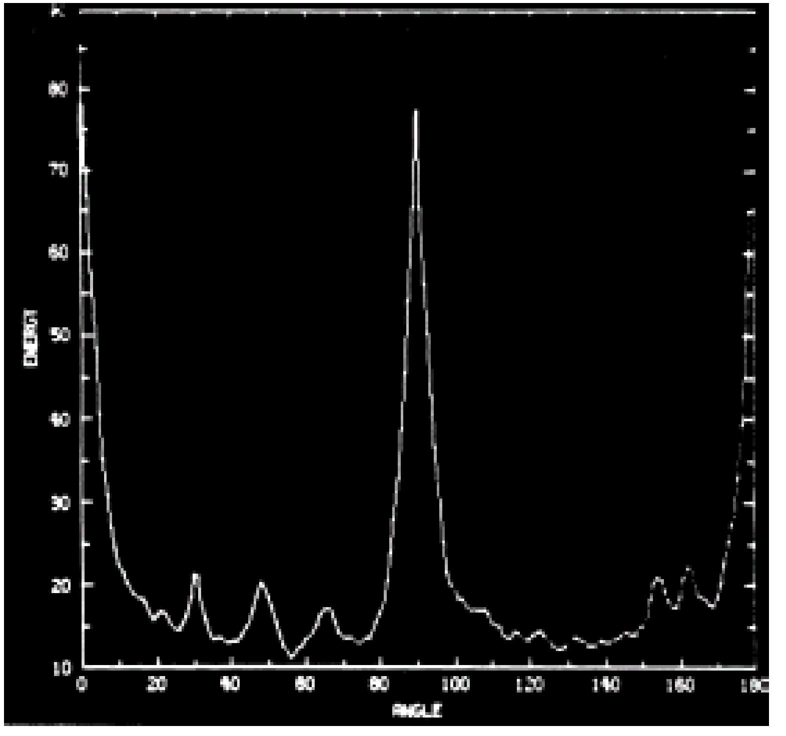


Plot of $S(\theta)$

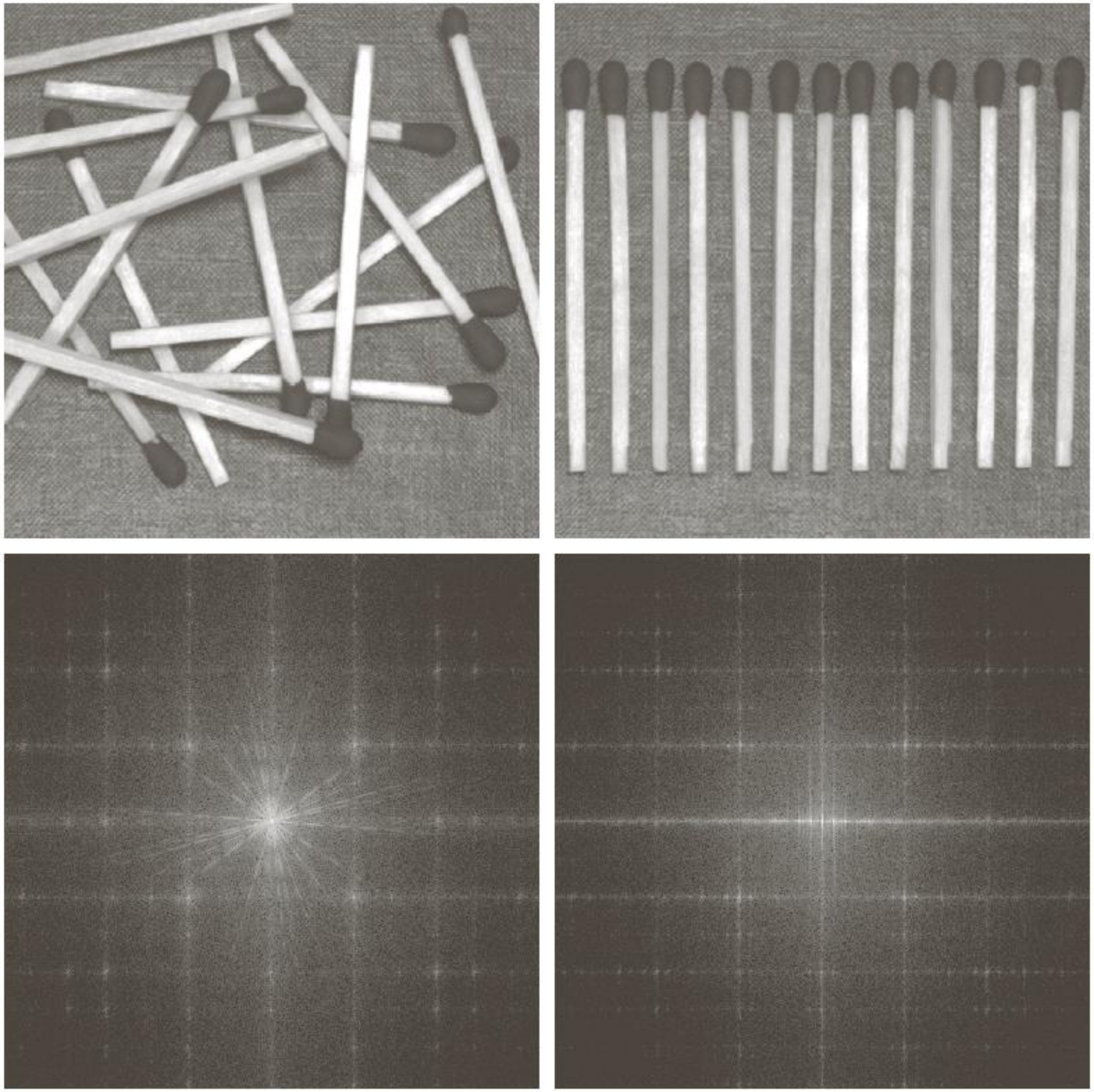
Spectral techniques: Fourier transform (example)



Another image showing periodic texture

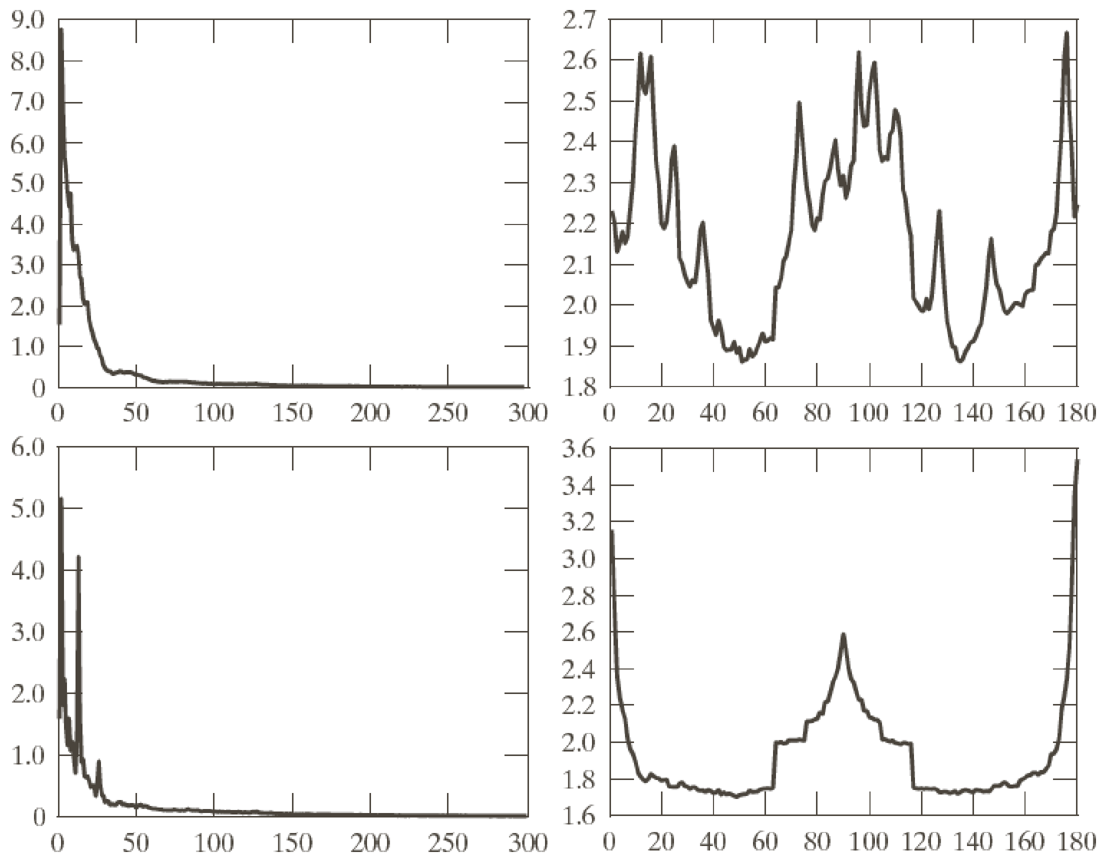


Plot of $S(\theta)$



a b
c d

FIGURE 11.35
(a) and (b) Images of random and ordered objects. (c) and (d) Corresponding Fourier spectra. All images are of size 600×600 pixels.



a b
c d

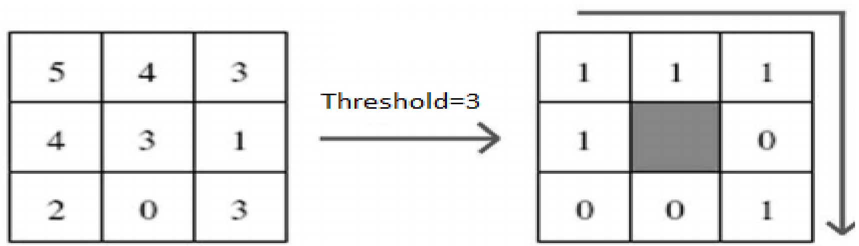
FIGURE 11.36 Plots of (a) $S(r)$ and (b) $S(\theta)$ for Fig. 11.35(a). (c) and (d) are plots of $S(r)$ and $S(\theta)$ for Fig. 11.35(b). All vertical axes are $\times 10^5$.

Local Binary Patterns (LBP)

LBP

- Mainly designed for monochrome still images
  - Have been extended for color (multi channel)
  - Videos …
- Were introduced by Ojala et al*. "A comparative study of texture measures with classification based on feature distributions". Pattern Recognition.* ***29****(1), 51–59* (**1996**)

- It is assumed that a texture has locally two complementary aspects, a pattern and its strength
- local binary pattern operator works in a 3×3 pixel



- The pixels in this block are
  - thresholded by its center pixel value,
  - multiplied by powers of two (Decimal)
  - then summed to obtain a label for the center pixel
  - 256 different labels

1. Local Binary Pattern (LBP)

- **Description of pixels neighbourhood**
- **Binary short code to describe neighbourhood**
- **Operates by taking difference of central pixel with neighbouring pixels**
- **Mathematically**

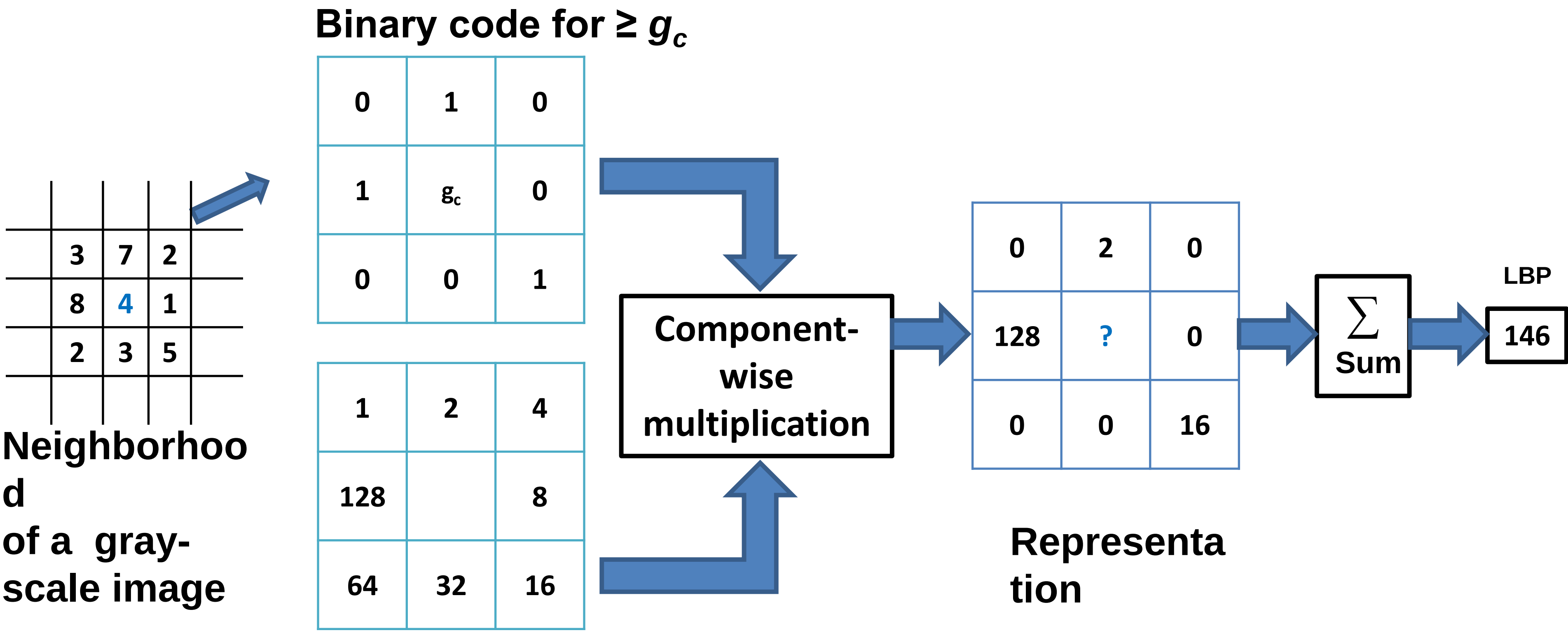
$$LBP_{R,P} = \sum_{p=0}^{P-1} s(g_p - g_c).2^p$$

where,
neighborhood pixels (g_p) in each block
is thresholded by its center pixel value (g_c)
$p \rightarrow$ sampling points (e.g., $p = 0, 1, \ldots, 7$ for a 3x3 cell, where P = 8)
$R \rightarrow$ radius (for 3x3 cell, it is 1).
Coordinates of "g_c"is (0,0) and of "g_p" is (x + Rcos(2π p/P), y − Rsin(2π p/P))

Binary threshold function s(x) is

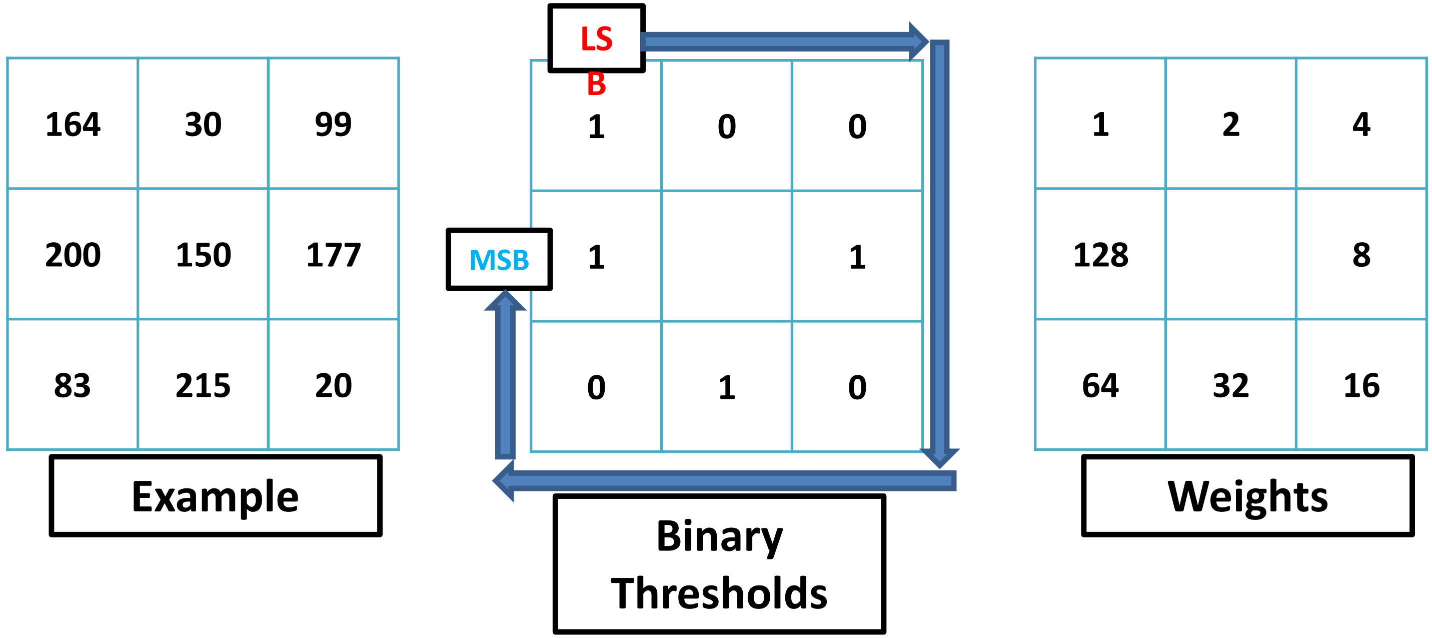
$$\mathbf{s}(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Computation of Local Binary Pattern



Example of how the *LBP operator* works

Computation of LBP



Binary Pattern:	1 (MSB)	0	1	0	1	0	0	1 (LSB)

Code/Weight (2^p):	1 x 2^7	0x 2^6	1 x 2^5	0x 2^4	1x 2^3	0x 2^2	0x 2^1	1 x 2^0
	= 128	= 0	= 32	= 0	= 8	= 0	= 0	= 1

LBP:	1 + 0 + 0 + 8 + 0+ 32 + 0+ 128 = 169

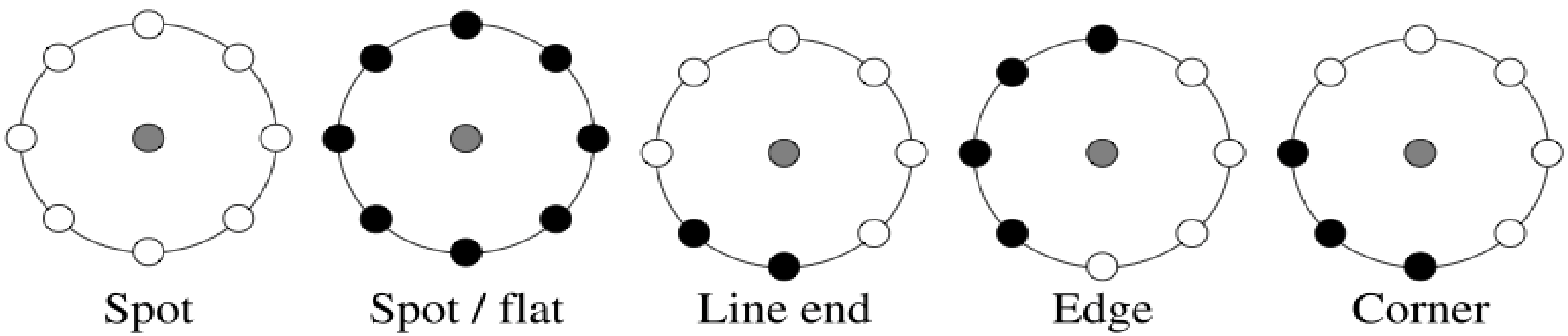
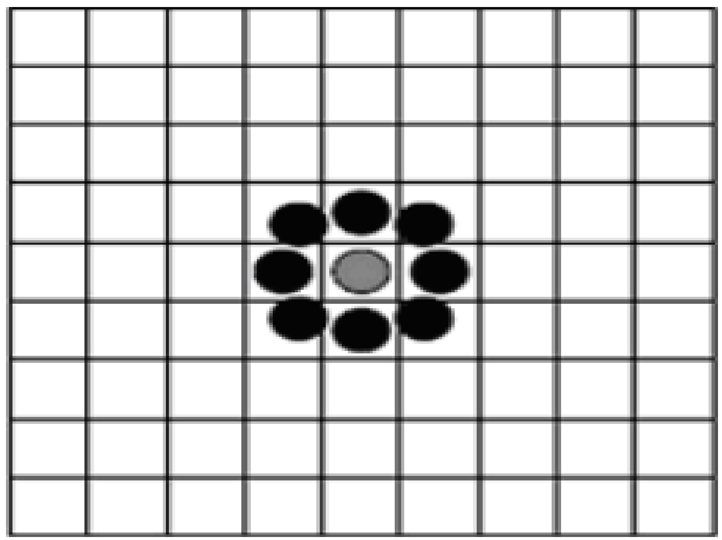


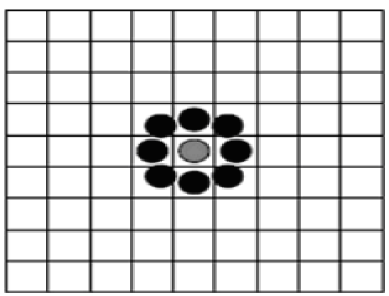
Fig. 2.3 Different texture primitives detected by the LBP

STANDARD LBP Filter

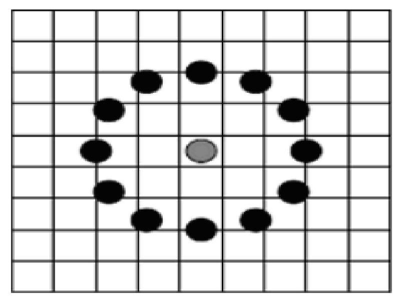


Advanced LBP $_{(P,R)}$

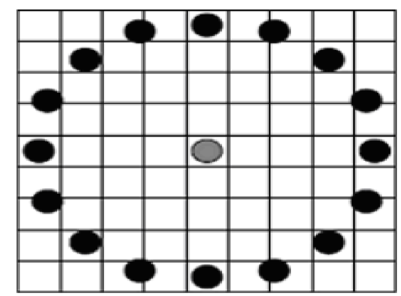
P = Pixels
R = Radius



LBP(8,1)



LBP(16,2)



LBP(20,4)

LBP Advantages and disadvantages

Advantages

- High discriminative power
- Computational simplicity
- Invariance to grayscale changes and
- Good performance.

Disadvantages

- Not invariant to rotations
- The size of the features increases exponentially with the number of neighbours which leads to an increase of computational complexity in terms of time and space
- The structural information captured by it is limited. Only pixel difference is used, magnitude information ignored.

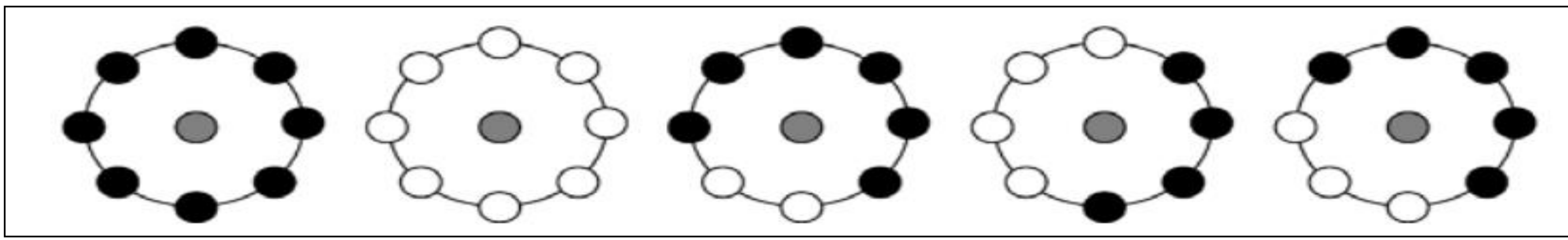
LBP: UNIFORM PATTERNS

uLBP

- Uniformity measure U (“pattern”) is the number of bitwise transitions from 0 to 1 or vice versa.
- A local binary pattern is called uniform if its uniformity measure is at most 2. i.e transitions between 0 and 1 ≤ 2

Example

- 00000000 (0 transitions)
- 01110000 (2 transitions)
- 11001111 (2 transitions)
- 11001001 (4 transitions)
- 01010011 (6 transitions)



uLBP

- In uniform LBP mapping there is a separate output label for each uniform pattern and all the non-uniform patterns are assigned to a single label.
- Why Omit non-uniform patterns?

Reasons for omitting non-uniform patterns

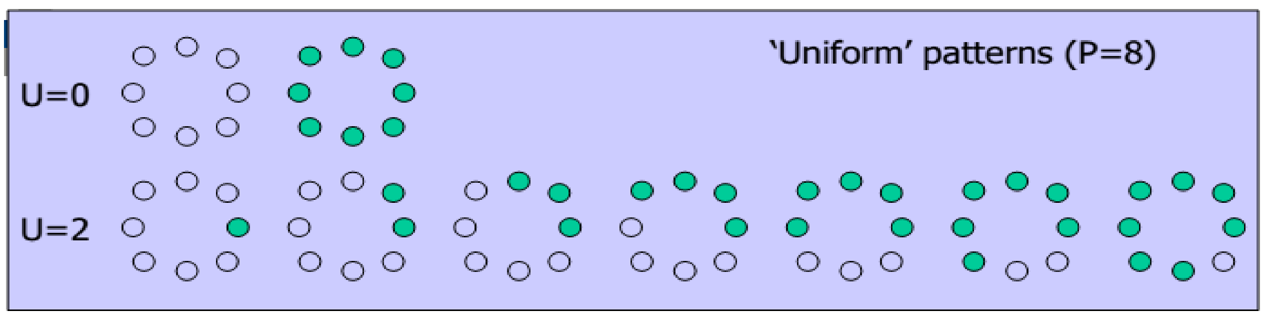
- most of the local binary patterns in natural images are uniform
- Ojala et al. noticed that in texture images, uLBP account for
  - 90% of all patterns using the (8,1)
  - 70% in the (16, 2) neighborhood.
- Facial images
  - 90.6% of the patterns in the (8, 1)
  - 85.2% of the patterns in the (8, 2)

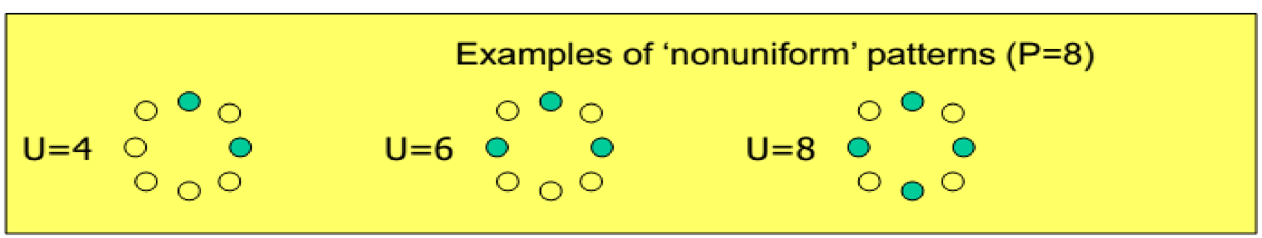
- **Uniform Value can be found using eq. below**

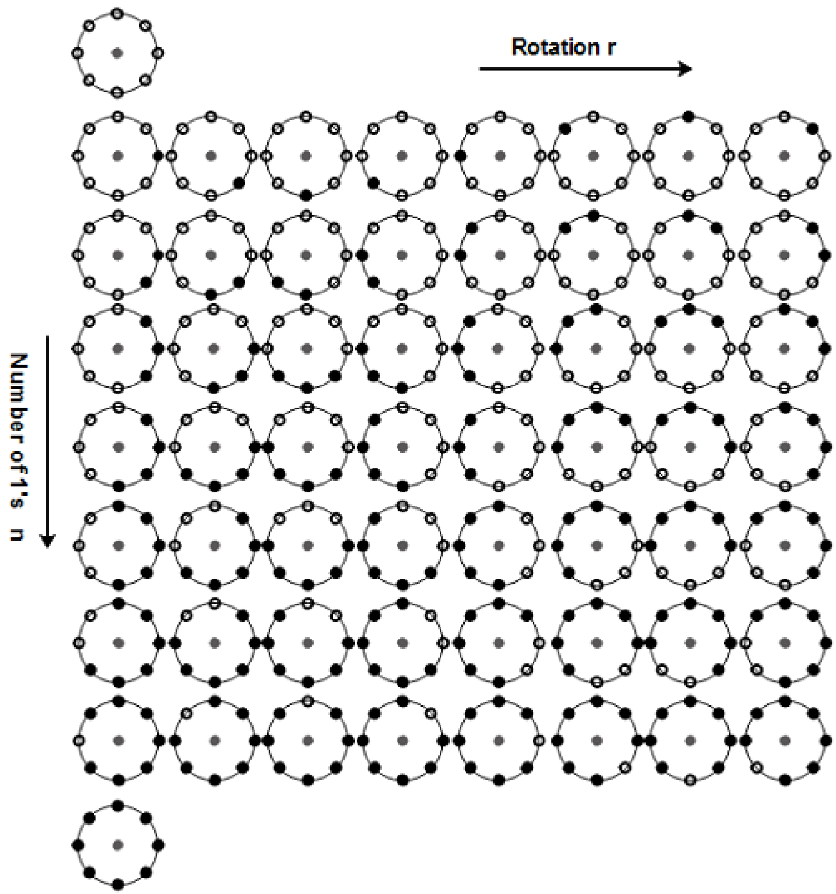
$$U\left(LBP_{P,R}\right) = \left|s(g_0 - g_C) - s(g_{P-1} - g_C)\right| + \sum_{p=1}^{P-1} \left|s(g_p - g_C) - s(g_{p-1} - g_C)\right|$$

- **If U≤2 it is uniform else non-uniform LBP**
- **Uniform LBP has P*(P-1)+2 output values**

Examples







- **A total of 58 binary patterns for (8, 1) neighbourhood**
- **‘r’ and ‘n’ shows rotation and No. of 1s respectively**

Advantages:

- **considers only the smooth patterns that account for the majority ((90%) for (8,1) and (70%) for (16, 2) neighbourhood) of the total binary patterns**
- **Only ‘‘uniform’’ patterns are fundamental patterns of local image texture.**
- **Lower dimensionality of features**

Disadvantages:

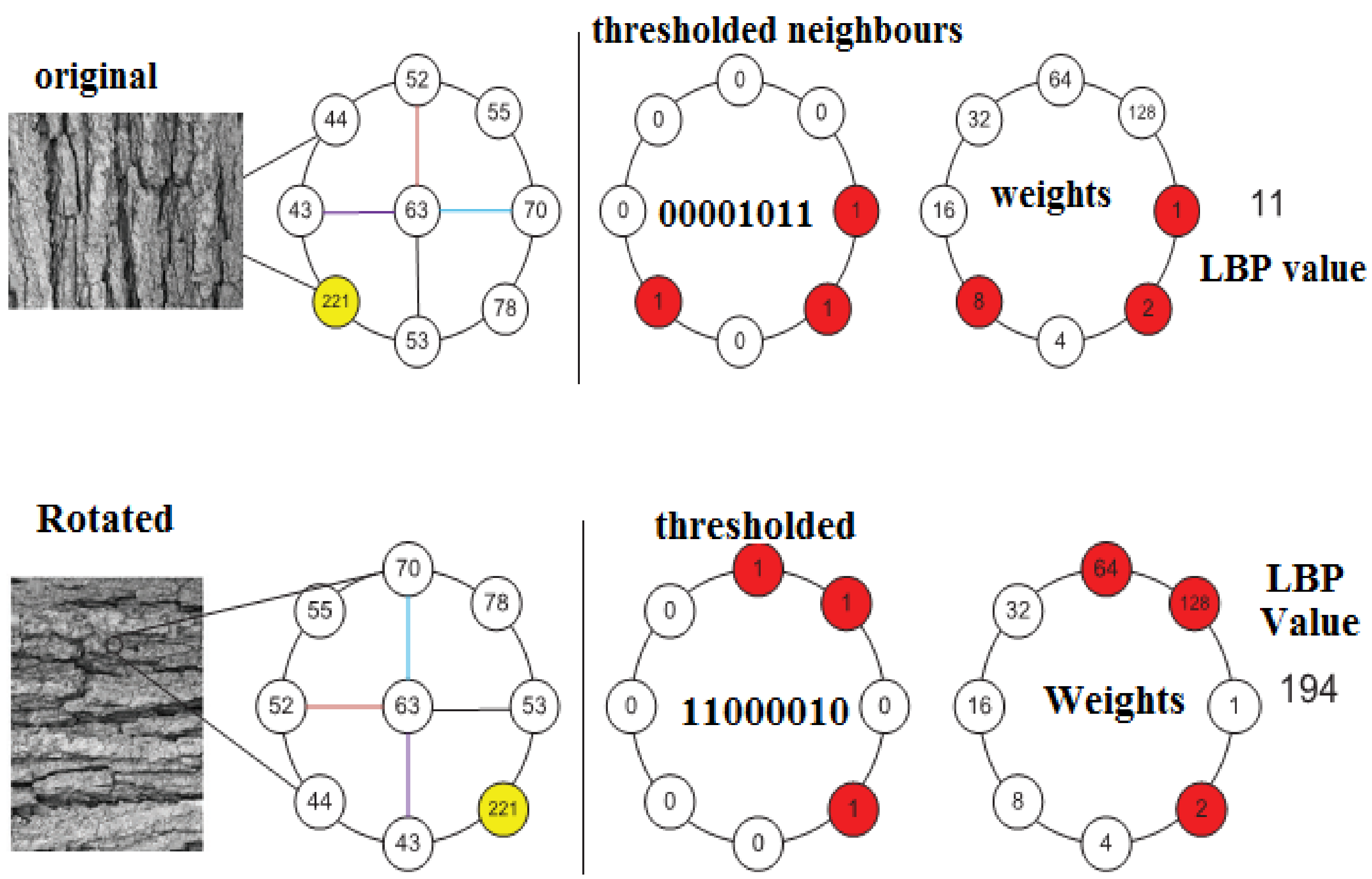
- **No rotation Invariant**

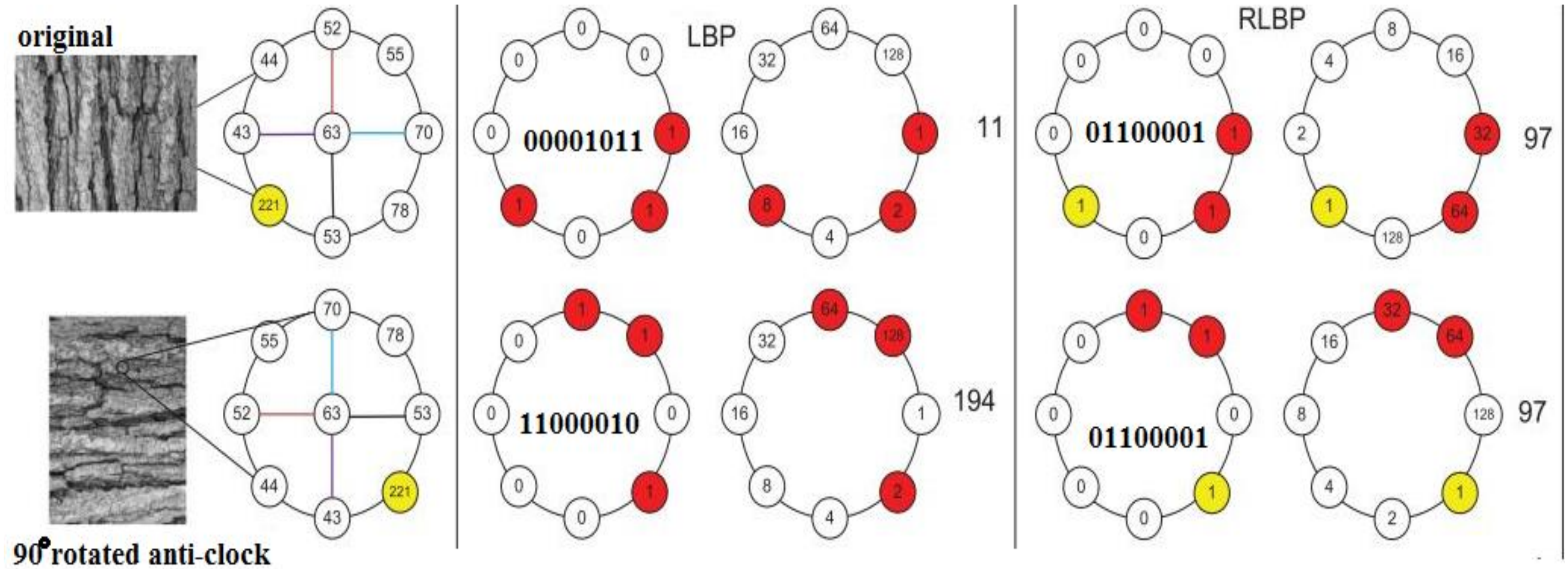
3. Rotated LBP

- **Problems in RILBP are resolved by RLBP**
- **Circularly shifting the weights of LBP operator**
- **Utilize magnitude of difference to find dominant direction in neighbourhood**
- **Dominant direction is the maximum difference of neighbouring pixels from central pixel**
- **Dominant direction is set as reference**

$$\mathbf{D} = \boldsymbol{argmax}_{\mathrm{p} \in (0,1,\ldots \mathrm{P}-1)} |\mathrm{g_p} - \mathrm{g_c}| \qquad \text{(Dominant Direction)}$$

$$\mathbf{RLBP_{R,P}} = \sum_{\mathbf{p=0}}^{\mathbf{P-1}} \mathbf{s}(\mathbf{g_P} - \mathbf{g_C}).\, \mathbf{2}^{\mathbf{mod(p-D,P)}} \qquad \text{(Rotated LBP)}$$


original
52
44
55
43
63
70
221
78
53
thresholded neighbours
0
0
0
0
00001011
1
1
0
1
64
32
128
16
weights
1
8
4
2
11
LBP value
Rotated
70
55
78
52
63
53
44
43
221
thresholded
1
0
1
0
11000010
0
0
0
1
64
32
128
LBP
Value
16
Weights
1
194
8
4
2



Advantages

- **Invariant to rotation**
- **High discriminative power**

Disadvantages

- **Large feature vector size**
- **Computational complexity**

Acknowledgements

Material in these slides has been taken from, the following resources

- Digital Image Processing", Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- Computer Vision for Computer Graphics, Mark Borg